

Niklas Schweiger

M.Sc. Student & Research Assistant · TU Munich

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Master's student at TU Munich and research assistant in Prof. Daniel Cremers' Computer Vision & AI group. Work focuses on inference-time alignment of diffusion and flow models. First-author paper accepted to ECCV 2026.

EDUCATION	M.Sc. Robotics, Cognition, Intelligence Technical University of Munich (TUM), Germany • Current GPA: 1.6 (German scale) • Focus: Machine learning, generative models, inference-time optimization • Exchange: Chalmers University of Technology, Gothenburg, Sweden (Sep 2024 – Jan 2025) B.Sc. Electrical Engineering & Information Technology Technical University of Munich (TUM), Germany • Final GPA: 2.5 (German scale) • Specialization: Artificial Intelligence and Machine Learning	Oct 2023 – Present Oct 2020 – Sep 2023
EXPERIENCE	Research Assistant (HiWi) Chair for Computer Vision & Artificial Intelligence (CVAI), TU Munich • Developed Trust-Region Noise Search (TRS), a black-box method for aligning diffusion and flow models without fine-tuning or gradient access. • Evaluated across text-to-image, molecule, and protein design against multiple baselines. • First-author paper accepted at ECCV 2026. Internship – AI in Industrial Production Siemens AG Developed a 3D feature matching system using CNN embeddings to automate part retrieval in manufacturing databases, bridging deep learning with industrial applications.	Mar 2026 – Present 2023
PUBLICATIONS	Schweiger, N., Ram, K., & Cremers, D. (2026). Trust-Region Noise Search for Black-Box Alignment of Diffusion and Flow Models. <i>European Conference on Computer Vision (ECCV), 2026. Also presented at ReALM-GEN Workshop @ ICLR 2026.</i>	
AWARDS & ACHIEVEMENTS	2nd Place – TUM.ai Makeathon Built <i>Caire</i>, an app designed to overcome language barriers between nurses and patients using AI to extract medical information from natural speech. Awarded a €2,000 prize.	Apr 2023
SKILLS & INTERESTS	Programming Python (Advanced), PyTorch (Advanced), C/C++ (Intermediate), Matlab (Intermediate), LaTeX Languages German (Native), English (C1 – Professional working proficiency) Interests Football, Strength training & running, Guitar (10+ years), AI & Natural sciences	